

Topic 6: Estimation

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Statistics (III) Midterm

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Core Questions

- What makes an estimator a “good” estimator?
- When population parameters are unknown, how do we estimate them from samples?
- Why does the quality of an estimator directly affect SPC?

Why Is Estimation Important in SPC?

In SPC, the in-control state of a process is usually described by baseline parameters such as μ_0 , σ_0

In practice, they are usually **unknown** and must be estimated from Phase I data

Key points:

- If $\hat{\mu}_0$ is estimated incorrectly, the center line will shift
- If $\hat{\sigma}_0$ is estimated incorrectly, the control limits will become too wide or too narrow
- Incorrect limits may lead to:
 - Too many false alarms
 - Delayed detection of abnormal changes

Therefore, SPC is not only about drawing charts. The first question is:

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How good is the estimator?

Estimators and Sampling Distributions

Suppose

$$X_1, \dots, X_n \stackrel{iid}{\sim} F_\theta$$

where θ is an unknown parameter

An **estimator** is a function of the sample:

$$\hat{\theta} = T(X_1, \dots, X_n)$$

Since $X_1, \dots, X_n$ are random, $\hat{\theta}$ is also a random variable

Therefore, we should not only ask “what value do we get?” , but also ask:

- What is $E(\hat{\theta})$?
- What is $Var(\hat{\theta})$?
- Does $\hat{\theta}$ converge to θ as $n \rightarrow \infty$?

Bias: Does the Estimator Deviate from the True Value?

Definition:

$$\text{Bias}(\hat{\theta}, \theta) = E(\hat{\theta}) - \theta$$

If

$$E(\hat{\theta}) = \theta, \quad \forall \theta$$

then $\hat{\theta}$ is called an **unbiased estimator**

Interpretation: Bias measures whether the estimator tends to overestimate or underestimate the true value on average

Example: Suppose $E(X_i) = \mu$ and $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$. Then $E(\bar{X}) = \mu$

Therefore, $\bar{X}$ is an unbiased estimator of μ

MSE: Unbiasedness Is Not Enough

Mean Squared Error is defined as

$$MSE(\hat{\theta}, \theta) = E[(\hat{\theta} - \theta)^2]$$

It can be **decomposed** into

$$MSE(\hat{\theta}, \theta) = Var(\hat{\theta}) + Bias^2(\hat{\theta}, \theta)$$

Key message:

- An estimator can be unbiased but still have very large variance
- A slightly biased estimator may have a much smaller MSE

Example

Goal: Estimate μ , assuming $X_1, \dots, X_n \sim N(\mu, \sigma^2)$

Consider two estimators: $\hat{\mu}_1 = \bar{X}$, $\hat{\mu}_2 = 0.8\bar{X}$

First estimator: $E(\hat{\mu}_1) = \mu$, $Var(\hat{\mu}_1) = \frac{\sigma^2}{n}$, then $MSE(\hat{\mu}_1) = \frac{\sigma^2}{n}$

Second estimator:

$$E(\hat{\mu}_2) = 0.8\mu, \quad Var(\hat{\mu}_2) = 0.8^2 Var(\bar{X}) = 0.64 \frac{\sigma^2}{n}$$

$$Bias(\hat{\mu}_2, \mu) = 0.8\mu - \mu = -0.2\mu,$$

$$\text{Then, } MSE(\hat{\mu}_2) = 0.64 \frac{\sigma^2}{n} + (0.2\mu)^2$$

Conclusion: A biased estimator may still have a smaller MSE if its variance is less

Consistency: Does the Estimator Approach the True Value as the Sample Size Increases?

Definition:

$$\hat{\theta}_n \xrightarrow{P} \theta$$

then $\hat{\theta}_n$ is called a **consistent estimator** of θ

That is
$$\lim_{n \rightarrow \infty} P(|\hat{\theta}_n - \theta| > \varepsilon) = 0, \quad \forall \varepsilon > 0$$

Interpretation:

- The estimator may have a large error when the sample size is small
- The estimator should become increasingly stable and closer to the true value when the sample size increases

A Common Sufficient Condition: $E(\hat{\theta}_n) \rightarrow \theta$ and $Var(\hat{\theta}_n) \rightarrow 0$

From Sufficient Condition to Mean Square Consistency

If $E(\hat{\theta}_n) \rightarrow \theta$ and $Var(\hat{\theta}_n) \rightarrow 0$
then

$$E\left[(\hat{\theta}_n - \theta)^2\right] = Var(\hat{\theta}_n) + \left(E(\hat{\theta}_n) - \theta\right)^2 \rightarrow 0$$

We call that **L2 (Mean Square) Consistency**:

$$\hat{\theta}_n \xrightarrow{L^2} \theta$$

Strength (Strongest to Weakest):

$$\hat{\theta}_n \xrightarrow{a.s.} \theta \quad \text{or} \quad \hat{\theta}_n \xrightarrow{L^2} \theta \quad \Rightarrow \quad \hat{\theta}_n \xrightarrow{P} \theta \quad \Rightarrow \quad \hat{\theta}_n \xrightarrow{d} \theta$$

Consistency of the Sample Mean

Suppose

$$X_1, \dots, X_n \stackrel{iid}{\sim} (\mu, \sigma^2)$$

and consider the sample mean

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

We have

$$E(\bar{X}) = \mu, \quad \text{Var}(\bar{X}) = \frac{\sigma^2}{n} \rightarrow 0$$

Then, by Chebyshev's inequality,

$$P(|\bar{X} - \mu| > \varepsilon) \leq \frac{\sigma^2/n}{\varepsilon^2} = \frac{\sigma^2}{n\varepsilon^2} \rightarrow 0, \quad \forall \varepsilon > 0$$

Therefore

$$\bar{X} \xrightarrow{P} \mu$$

Consistency of the Sample Variance

Consider the sample variance

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

We have

$$E(S^2) = \sigma^2, \quad \text{Var}(S^2) \rightarrow 0$$

Then, by Chebyshev's inequality,

$$P(|S^2 - \sigma^2| > \varepsilon) \leq \frac{E[(S^2 - \sigma^2)^2]}{\varepsilon^2} = \frac{MSE(S^2)}{\varepsilon^2}$$

Since $E(S^2) = \sigma^2$, S^2 is unbiased, so

$$P(|S^2 - \sigma^2| > \varepsilon) \leq \frac{\text{Var}(S^2)}{\varepsilon^2} \rightarrow 0, \quad \forall \varepsilon > 0$$

Therefore

$$S^2 \xrightarrow{P} \sigma^2$$

Example

Suppose

$$X_1, \dots, X_n \stackrel{iid}{\sim} \text{Bernoulli}(p)$$

where $P(X_i = 1) = p$, $P(X_i = 0) = 1 - p$, $E(X) = p$, $\text{Var}(X) = p(1 - p)$

Since $\hat{p} = \bar{X} = \frac{1}{n} \sum_{i=1}^n X_i \Rightarrow E(\hat{p}) = E(\frac{1}{n} \sum_{i=1}^n X_i) = \frac{1}{n} * n * p = p$

$$\text{Var}(\hat{p}) = \frac{1}{n^2} * n * p(1 - p) = \frac{p(1-p)}{n}$$

Therefore $E(\hat{p}) \rightarrow p$ and $\text{Var}(\hat{p}) = \frac{p(1-p)}{n} \rightarrow 0$ as $n \rightarrow \infty$

So

$$\hat{p} \xrightarrow{P} p \Rightarrow \hat{p} \text{ is consistent}$$

From Estimation to SPC: The Core of Phase I

In Phase I, we first establish an in-control baseline

For an $\bar{X}$ -chart, if μ_0 and σ_0 are known, the control limits are

$$UCL = \mu_0 + z_{1-\alpha/2} \frac{\sigma_0}{\sqrt{n}}, \quad LCL = \mu_0 - z_{1-\alpha/2} \frac{\sigma_0}{\sqrt{n}}$$

Usually, we replace them with estimators $\hat{\mu}_0$ and $\hat{\sigma}_0$

Thus, the actual control limits become

$$\widehat{UCL} = \hat{\mu}_0 + z_{1-\alpha/2} \frac{\hat{\sigma}_0}{\sqrt{n}}, \quad \widehat{LCL} = \hat{\mu}_0 - z_{1-\alpha/2} \frac{\hat{\sigma}_0}{\sqrt{n}}$$

Potential issues:

- If $\hat{\mu}_0$ is biased, the center line shifts.
- If $\hat{\sigma}_0$ has large variance, the control limits become unstable.
- If the estimators are not consistent, we cannot establish a reliable baseline.

How Estimator Affects Monitoring: Example of a p-Chart

Suppose each subgroup has sample size n , and the defect rate is p_0 . Then

$$\hat{p} = \frac{X}{n}, \quad X \sim \text{Binomial}(n, p_0)$$

In Phase I, we usually use $\bar{p} = \frac{1}{m} \sum_{i=1}^m \hat{p}_i$ to estimate p_0 .

The control limits of the p-chart are often written as

$$UCL = \bar{p} + z\sqrt{\frac{\bar{p}(1 - \bar{p})}{n}}, \quad LCL = \bar{p} - z\sqrt{\frac{\bar{p}(1 - \bar{p})}{n}}$$

Conclusion

(1) A good estimator should not be judged only by unbiasedness

- $MSE = Var + Bias^2$

(2) Consistency is a requirement for long-run reliability

- $\hat{\theta}_n \xrightarrow{P} \theta$

(3) In SPC, estimation is the foundation

- Phase I requires estimation of μ_0 and σ_0 in order to determine control limits
- Control limits determine false alarm rates and detection power

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If the baseline estimation is unreliable,
then the entire SPC monitoring system becomes unreliable.

Thank You